E5 base

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Abstracts

Conneau et al. (2020) (https://aclanthology.org/2020.acl-main.747)
The paper shows that pretraining multilingual models at large scale gives big gains on

cross-lingual transfer tasks. Its XLM-R model beats multilingual BERT on XNLI, MLQA, and NER, with especially strong gains for low-resource languages such as Swahili and Urdu. The authors also report that multilingual modeling can stay competitive with strong monolingual models on GLUE and XNLI.

Wang et al. (2022) (https://arxiv.org/abs/2212.03533)
The paper presents E5, a family of general-purpose text embeddings trained contrastively on a curated, large-scale set of text pairs (CCPairs), for retrieval, clustering, and classification. It is evaluated on 56 BEIR and MTEB datasets. Zero-shot, it is the first model to outperform BM25 on BEIR without labeled data, and fine-tuned, it achieves the best MTEB results, beating models with far more parameters.

Wang et al. (2024) (https://arxiv.org/abs/2402.05672)
This technical report describes the open-source multilingual E5 models in small, base, and large sizes. Training follows the English E5 recipe: contrastive pre-training on 1 billion multilingual text pairs, then fine-tuning on labeled datasets. It also introduces an instruction-tuned embedding model whose performance is on par with state-of-the-art English-only models of similar size.

Winata et al. (2024) (https://aclanthology.org/2024.findings-emnlp.155)
The authors note that how well multilingual language models work for semantic retrieval had not been thoroughly studied. They introduce MINERS, a benchmark covering bitext mining and classification via retrieval-augmented context, across more than 200 languages including low-resource, cross-lingual, and code-switching settings. They report that simply retrieving semantically similar embeddings is competitive with state-of-the-art approaches, with no fine-tuning needed.

Niklaus et al. (2024) (https://aclanthology.org/2024.acl-long.805)
The paper argues that specialized domains like law lack large, high-quality datasets. It releases MultiLegalPile, a 689GB corpus covering 24 languages and 17 jurisdictions, and pretrains multilingual and monolingual models on it. The multilingual models set a new state of the art on LEXTREME, and the English models on LexGLUE. The dataset, models, and code are released openly.

Lewis et al. (2020) (https://arxiv.org/abs/2005.11401)
The paper notes that large pretrained models store knowledge but struggle to access and update it precisely. It proposes retrieval-augmented generation (RAG), which combines a pretrained generator with a dense retrieval index over Wikipedia. RAG sets state-of-the-art results on three open-domain QA tasks and produces more specific, diverse, and factual text than a parametric-only baseline.

Es et al. (2024) (https://aclanthology.org/2024.eacl-demo.16)
The paper introduces RAGAs, a framework for reference-free evaluation of RAG pipelines. It addresses the several dimensions that matter: whether retrieval finds relevant and focused

passages, whether the LLM uses them faithfully, and the quality of the generation itself. Its metrics need no ground-truth human annotations, which speeds up evaluation cycles.

NLI

1. Five relevant NLI research papers

1. NLI models for evaluating faithfulness

Steen et al. (2023) — With a Little Push, NLI Models can Robustly and Efficiently Predict Faithfulness

Abstract summary: The researchers investigated the use of NLI models to determine whether generated text is supported by its source evidence. They found that NLI-based faithfulness evaluation can improve through task-specific data augmentation, consideration of both entailment and contradiction probabilities, and Monte Carlo dropout. Their findings demonstrate that an NLI model can serve as an efficient faithfulness evaluator when appropriately adapted to the task.

ACL Anthology

Relevance to CIVIL LEX: This is one of the strongest sources for your NLI component because your system evaluates whether generated legal explanations are supported by retrieved Civil Code provisions.

APA 7th edition:

Steen, J., Opitz, J., Frank, A., & Markert, K. (2023). With a little push, NLI models can robustly and efficiently predict faithfulness. Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), 914–924. https://doi.org/10.18653/v1/2023.acl-short.79

Read the full paper

2. Fine-grained NLI-based faithfulness evaluation

Zhang et al. (2024) — Fine-Grained Natural Language Inference Based Faithfulness Evaluation for Diverse Summarisation Tasks

Abstract summary: The study examined limitations in existing NLI-based approaches to evaluating generated summaries. The researchers proposed INFUSE, a method that adjusts the amount of source evidence considered and simplifies generated sentences into shorter

hypotheses. Their findings showed improved performance across diverse summarization tasks, highlighting the importance of how source evidence and generated claims are structured during entailment evaluation.

ACL Anthology

Relevance to CIVIL LEX: Legal explanations can contain several claims supported by different statutory articles. This study supports evaluating individual claims against relevant evidence rather than treating an entire answer as one undivided statement.

APA 7th edition:

Zhang, H., Xu, Y., & Perez-Beltrachini, L. (2024). Fine-grained natural language inference based faithfulness evaluation for diverse summarisation tasks. Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), 1701–1722. https://doi.org/10.18653/v1/2024.eacl-long.102

Read the full paper

3. Cross-lingual NLI and hallucination detection

Zhang and Perez-Beltrachini (2024) — Leveraging Entailment Judgements in Cross-Lingual Summarisation

Abstract summary: This study explored the use of cross-lingual NLI to identify generated summaries containing information unsupported by their source documents. The researchers investigated methods for detecting unfaithful content and improving model training to reduce hallucinations while retaining useful information. Their results support the use of entailment judgments to assess factual consistency in multilingual generation tasks.

ACL Anthology

Relevance to CIVIL LEX: This research is relevant when legal questions or explanations use Filipino, English, or mixed-language phrasing. However, cross-lingual performance does not automatically establish accuracy for Philippine legal terminology; your system still needs its own evaluation.

APA 7th edition:

Zhang, H., & Perez-Beltrachini, L. (2024). Leveraging entailment judgements in cross-lingual summarisation. Findings of the Association for Computational Linguistics: ACL 2024, 14481–14497. https://doi.org/10.18653/v1/2024.findings-acl.862

Read the full paper

4. Adversarial Natural Language Inference

Nie et al. (2020) — Adversarial NLI: A New Benchmark for Natural Language Understanding

Abstract summary: The authors introduced the Adversarial NLI (ANLI) benchmark, which was developed through an iterative process involving human annotators and machine-learning models. Their experiments demonstrated that adversarially constructed examples create a more difficult test of NLI systems than conventional benchmark examples. The study also showed that existing models remain vulnerable to challenging inference cases.

ACL Anthology

Relevance to CIVIL LEX: Legal questions often involve exceptions, conditions, and distinctions between similar statutory provisions. ANLI provides a useful foundation for understanding why an NLI model should be tested on difficult examples, not only straightforward entailment cases.

APA 7th edition:

Nie, Y., Williams, A., Dinan, E., Bansal, M., Weston, J., & Kiela, D. (2020). Adversarial NLI: A new benchmark for natural language understanding. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 4885–4901. https://doi.org/10.18653/v1/2020.acl-main.441

Read the full paper

5. NLI with faithful explanations

Kumar and Talukdar (2020) — NILE: Natural Language Inference with Faithful Natural Language Explanations

Abstract summary: The researchers introduced NILE, a method designed to produce NLI predictions alongside explanations that reflect the reasoning behind those predictions. They evaluated the method using automated and human assessments and emphasized that prediction accuracy alone is insufficient to establish the faithfulness of explanations. Their findings highlight the value of examining whether a model's decisions are genuinely supported by the evidence used to make them.

ACL Anthology

Relevance to CIVIL LEX: This study supports the importance of evidence-based explanations and transparent inference decisions in your legal RAG system. It is particularly useful when discussing why legal answers should be accompanied by relevant statutory citations.

APA 7th edition:

Kumar, S., & Talukdar, P. (2020). NILE: Natural language inference with faithful natural language explanations. Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics, 8730–8742. https://doi.org/10.18653/v1/2020.acl-main.771

Read the full paper